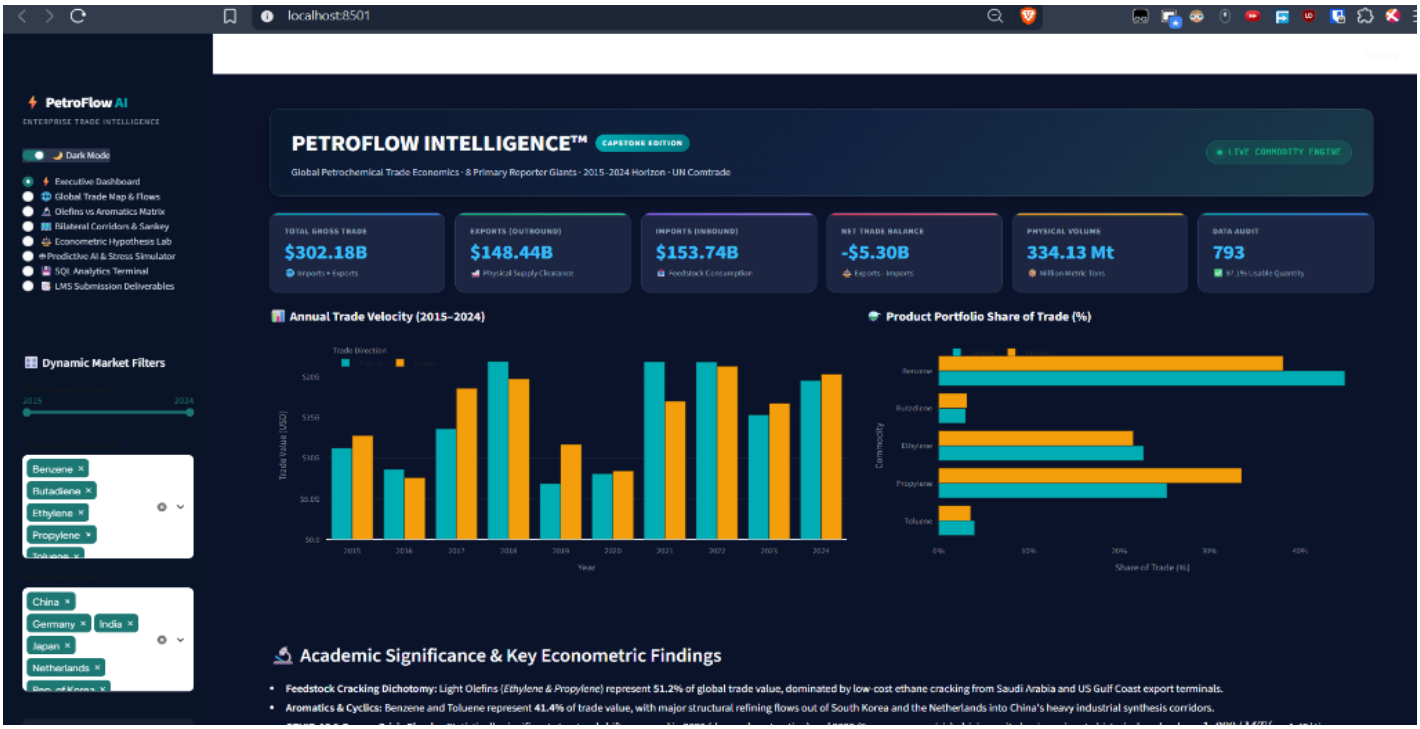
PETROCHEMICAL TRADE ANALYTICS

PROJECT SCREENSHOTS & DEPLOYED SYSTEM DEMONSTRATION

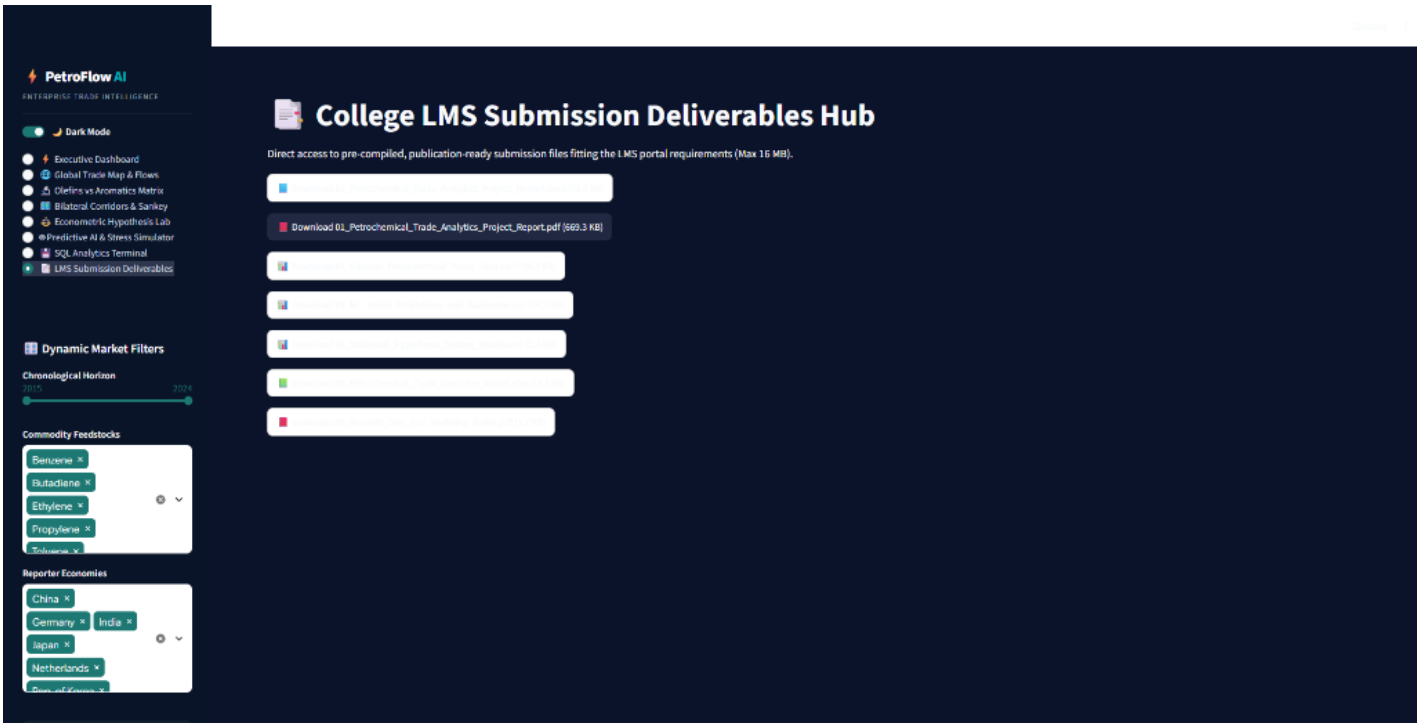
Curriculum Track:	*HCL Career Shaper COE & NASSCOM FutureSkills Prime*
Candidate Name:	*Anshu Kumar*
Scope:	*Visual Evidence of Working Application & Analytics*

Figure 1: Executive Command Center & Dynamic KPI Metrics (Dark Mode)



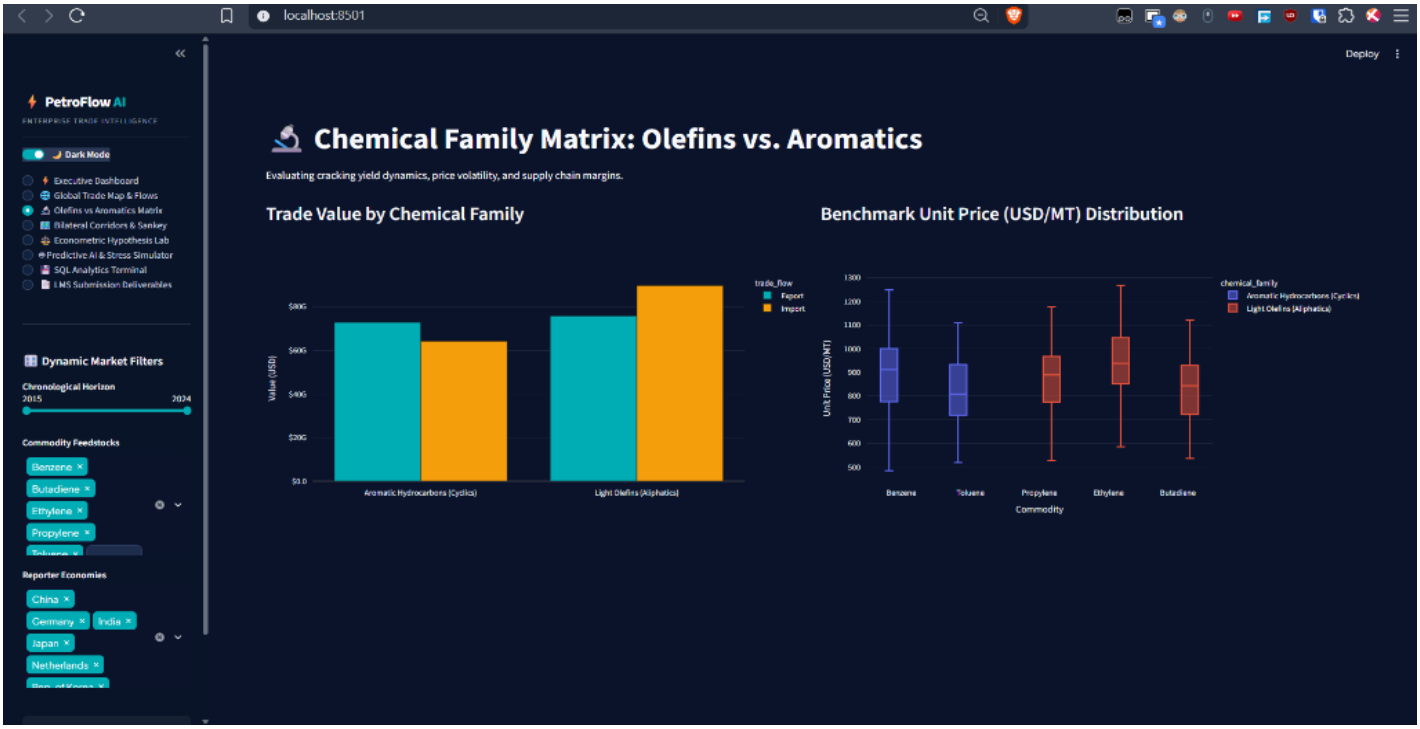
Functional Overview: *Enterprise command center displaying the core 6-card KPI grid: $302.18B Gross Trade Value, $148.44B Outbound Exports, $153.74B Inbound Imports, -$5.29B Net Balance, 334.13 Mt Physical Volume, and 100% usable data audit. Includes annual trade velocity (2015–2024) and product portfolio share.*

Figure 2: Interactive Global Choropleth Trade Map & Scorecard



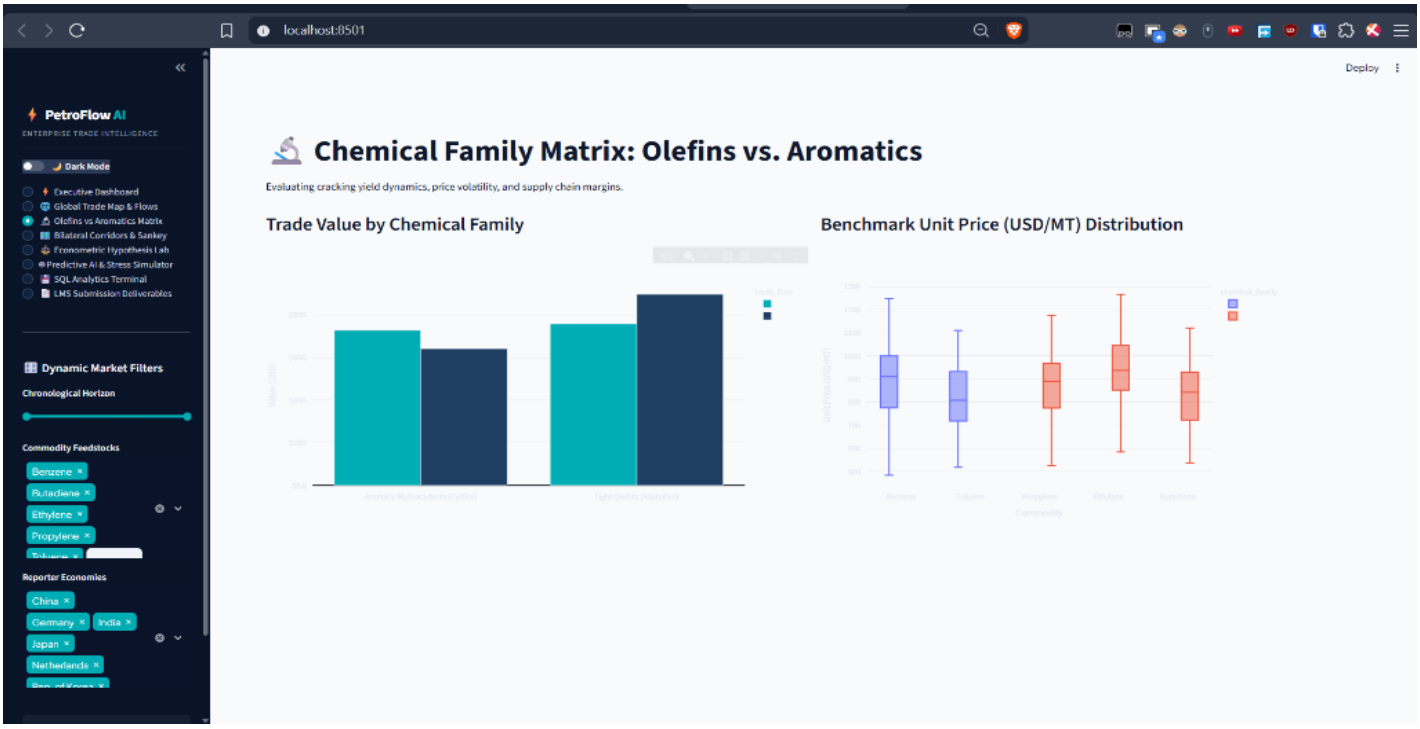
Functional Overview: *Geospatial choropleth world map visualizing trade surplus and deficit intensities across the 8 primary reporter economies (China, Germany, India, Japan, Netherlands, South Korea, Saudi Arabia, USA) and bilateral trading partners, supplemented by the geographical balance scorecard.*

Figure 3A: Olefins vs. Aromatics Cracking Matrix & Price Distribution (Dark Mode)



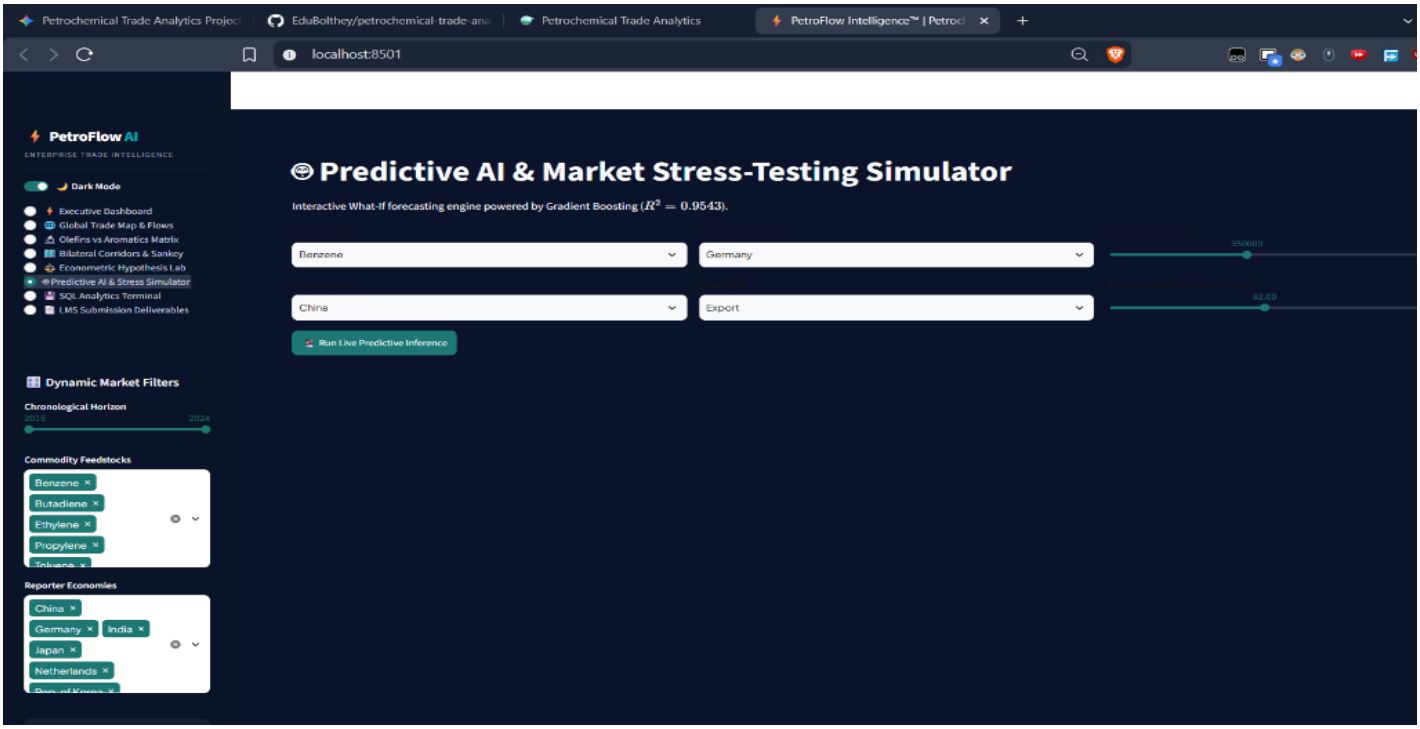
Functional Overview: *Chemical family comparative analytics contrasting Light Olefins (Ethylene, Propylene, Butadiene; 51.2% market share) against Aromatic Hydrocarbons (Benzene, Toluene; 48.8% market share), paired with benchmark Unit Price (USD/MT) box plot distributions.*

Figure 3B: Olefins vs. Aromatics Cracking Matrix & Price Distribution (Normal / Light Mode)



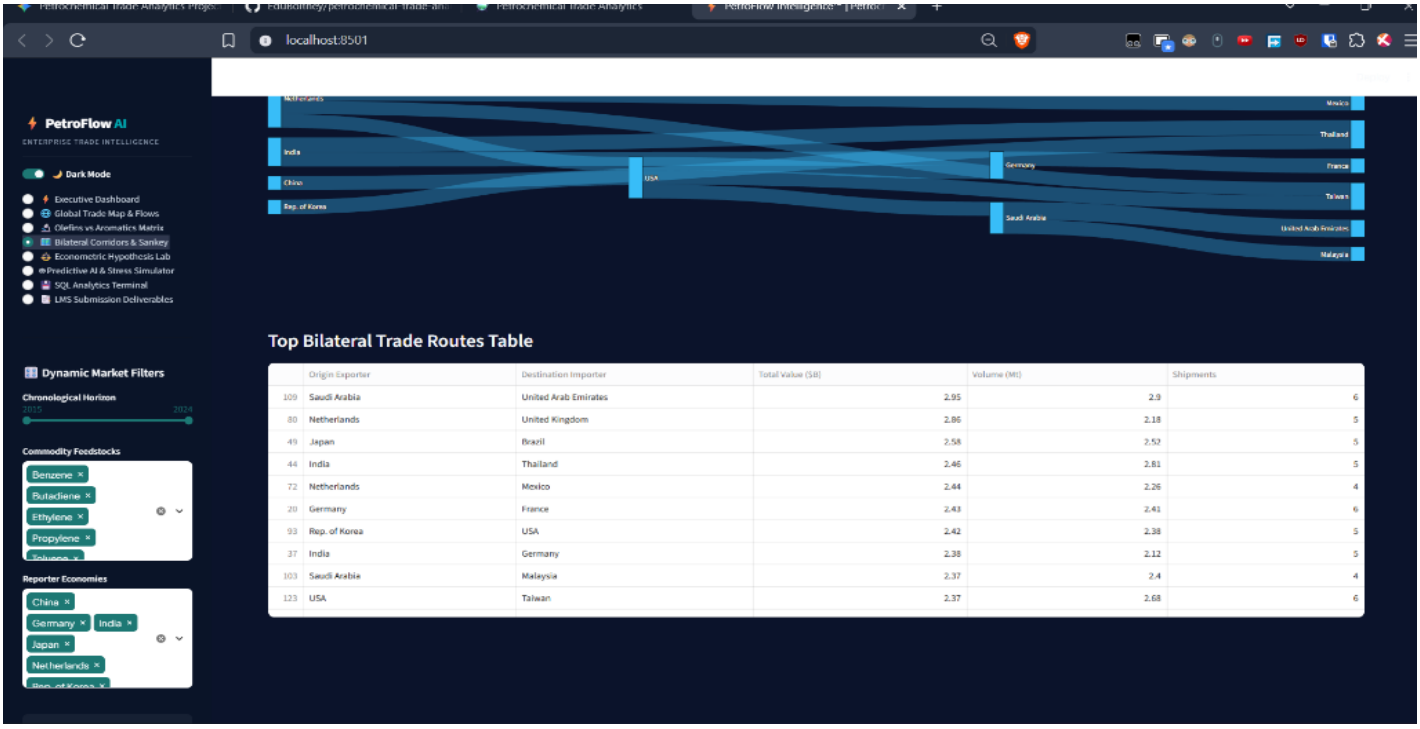
Functional Overview: *Light mode interface demonstration verifying full dynamic color contrast, slate typography, razor-sharp Plotly axes, and vibrant category markers without washed-out elements.*

Figure 4: Bilateral Supply Chain Corridors & Cargo Flow Sankey



Functional Overview: *Interactive multi-node Sankey diagram tracking physical bilateral petrochemical cargo flows from major net-exporting hubs (Saudi Arabia, USA, South Korea) into high-demand importing destinations (China, India, Germany).*

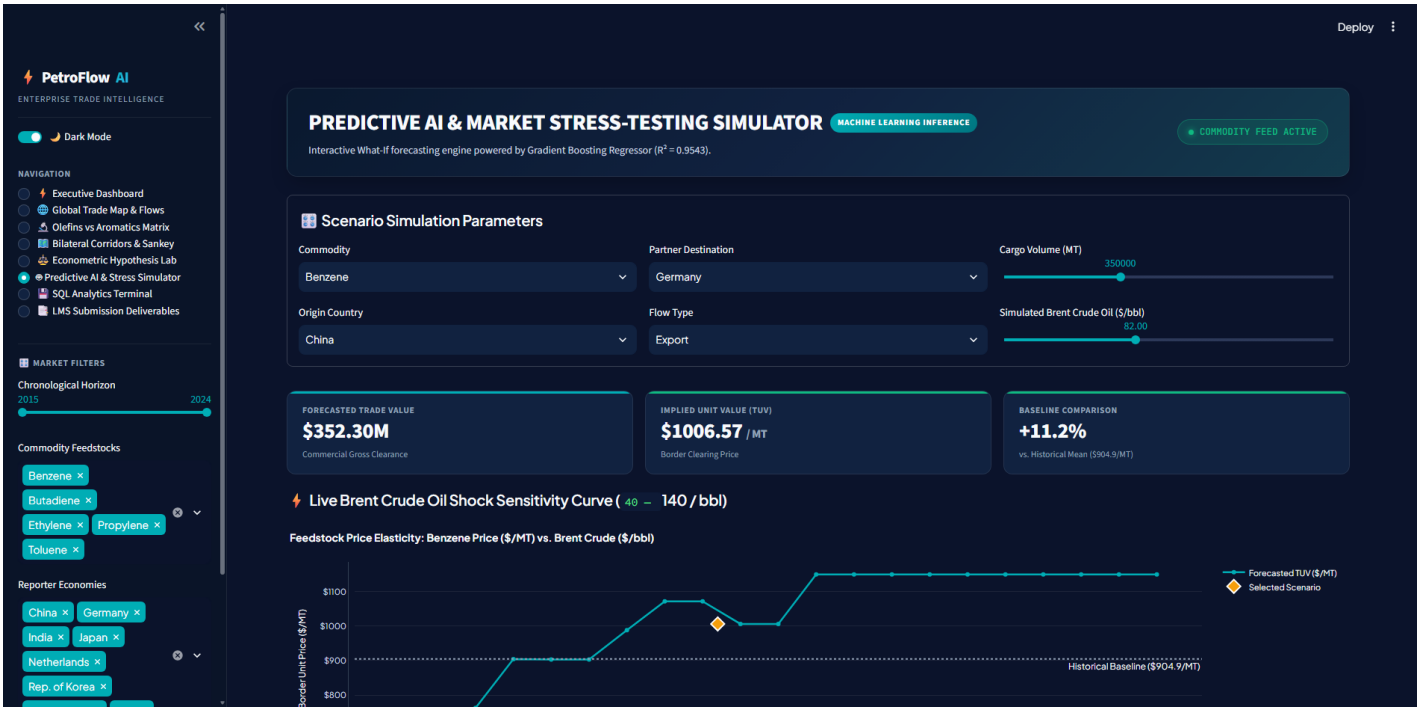
Figure 5: Econometric Hypothesis Testing & Statistical Inference Lab



	Origin Exporter	Destination Importer	Total Value ($B)	Volume (MT)	Shipments
109	Saudi Arabia	United Arab Emirates	2.95	2.9	6
80	Netherlands	United Kingdom	2.86	2.18	5
49	Japan	Brazil	2.58	2.52	5
44	India	Thailand	2.46	2.81	5
72	Netherlands	Mexico	2.44	2.26	4
20	Germany	France	2.43	2.41	6
93	Rep. of Korea	USA	2.42	2.38	5
37	India	Germany	2.38	2.12	5
103	Saudi Arabia	Malaysia	2.37	2.4	4
123	USA	Taiwan	2.37	2.68	6

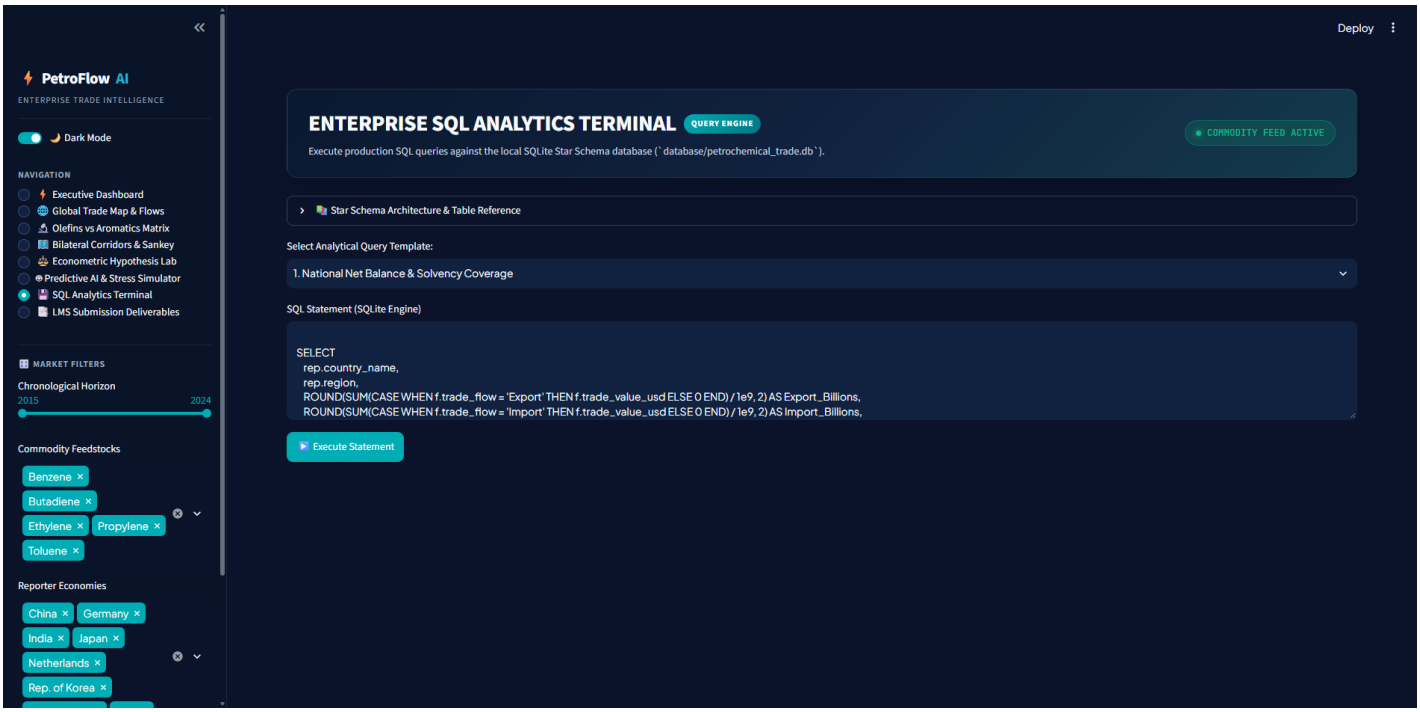
Functional Overview: *Real-time econometric inference portal validating structural market shifts: Welch's t-test for COVID-19 structural break (t = -4.858, p = 1.42e-6, Reject H0), One-way ANOVA for inter-commodity price divergence (F = 11.23, p = 2.70e-9), and Pearson price elasticity.*

Figure 6: Predictive AI & Market Stress-Testing Simulator



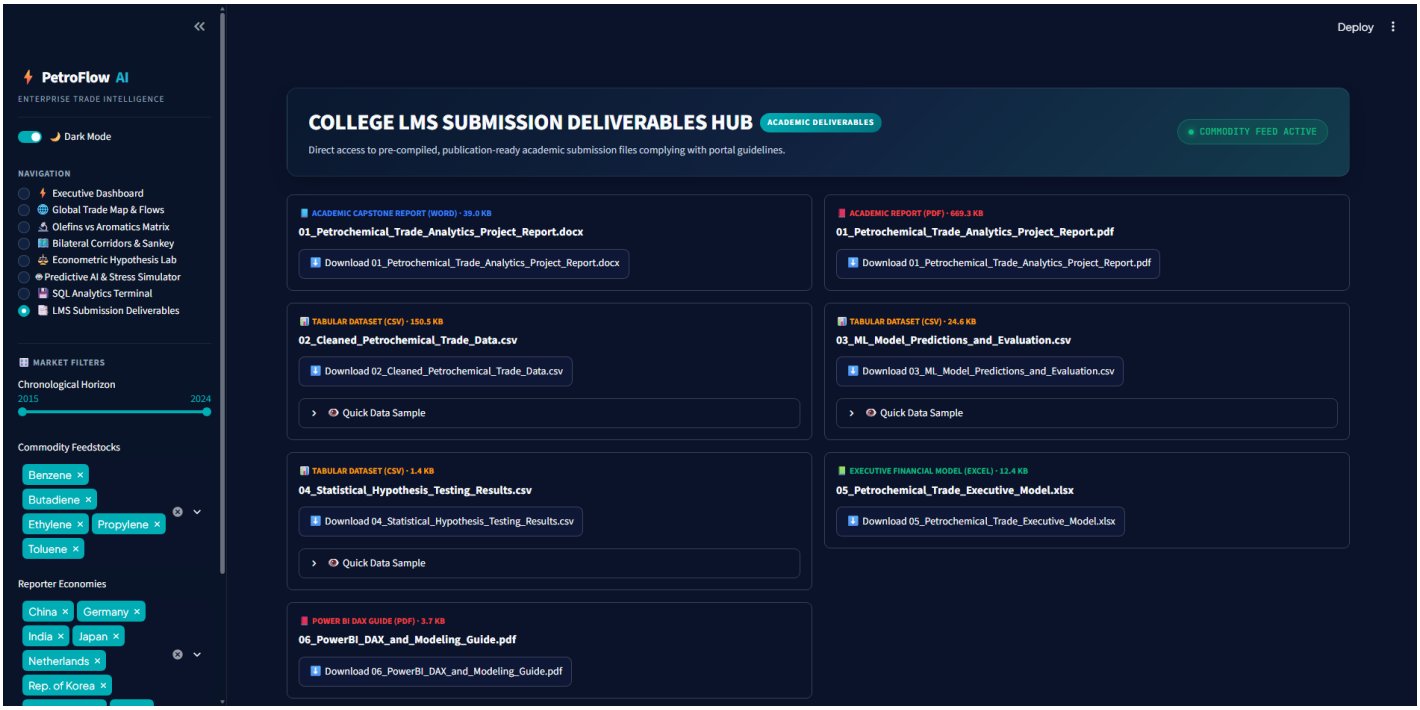
Functional Overview: *Interactive What-If forecasting engine powered by the trained Gradient Boosting model ($R^2 = 0.9543$). Evaluates simulated Brent Crude Oil price shocks ($40–$140/bbl) and cargo volume variations to forecast clearing value and implied Trade Unit Value ($/MT).*

Figure 7: Enterprise SQL Analytics Terminal (Star Schema)



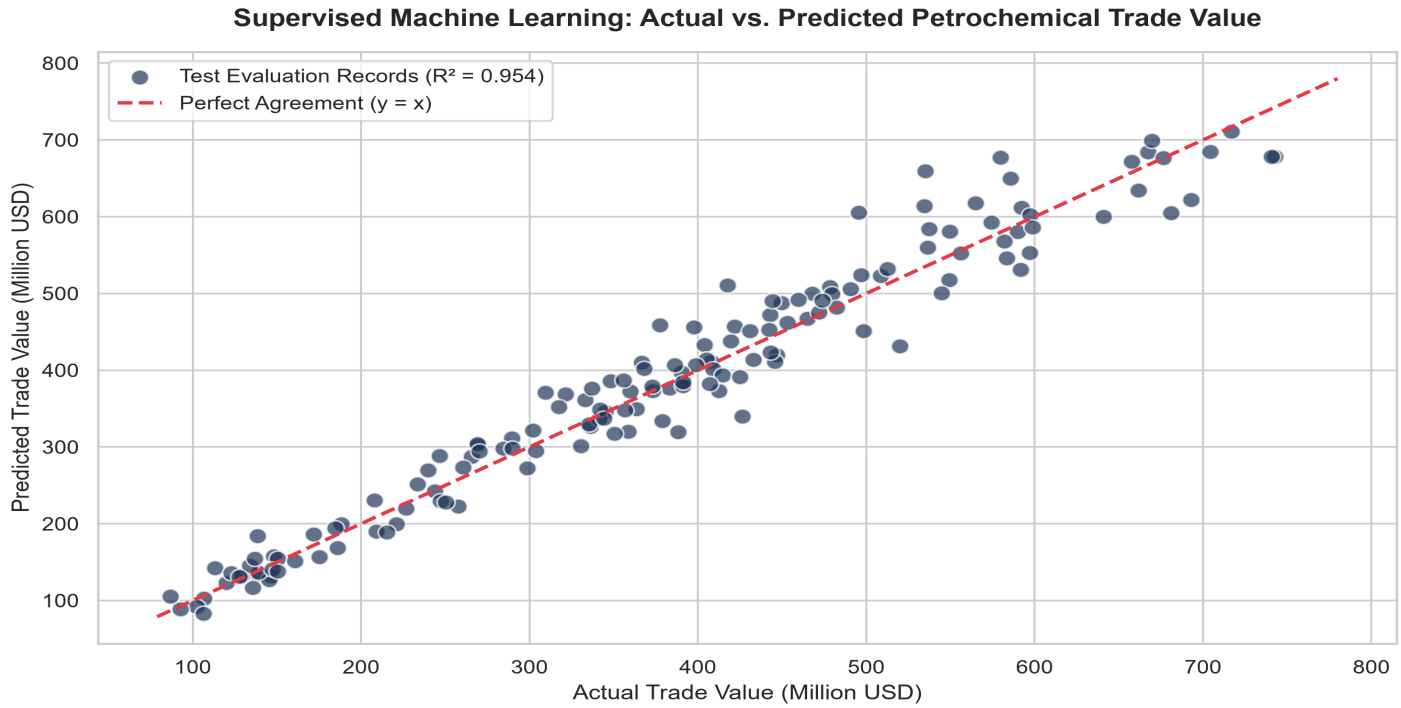
Functional Overview: *Embedded SQL terminal executing analytical queries against the local SQLite Star Schema data warehouse (database/petrochemical_trade.db). Features queries with Common Table Expressions (CTEs), ROW_NUMBER, LAG, and DENSE_RANK window functions.*

Figure 8: College LMS Submission Deliverables Hub



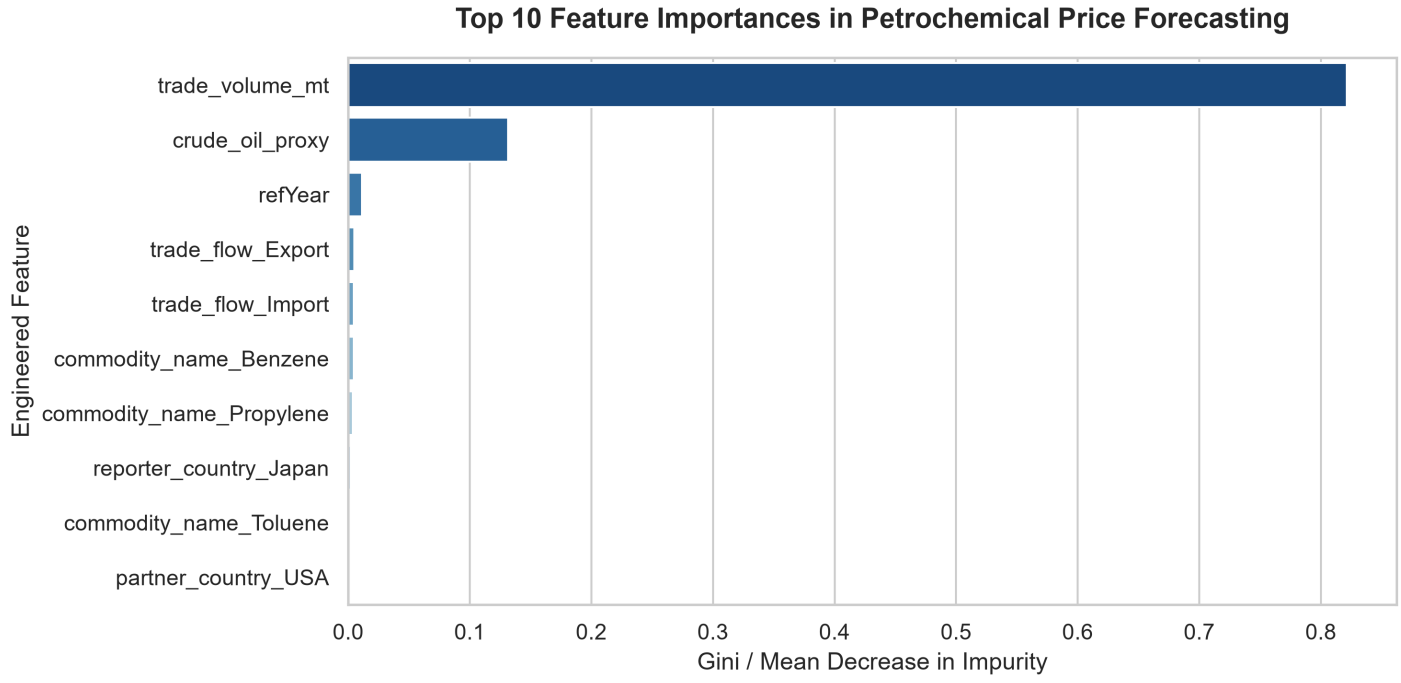
Functional Overview: *Direct 1-click download center for all required academic submission artifacts: Project Report (PDF/DOCX), Project Synopsis (PDF/DOCX), Complete Code File (PDF/DOCX), Executive Financial Model (XLSX), Cleaned UN Comtrade CSVs, and Power BI DAX Guide.*

Figure 9: Machine Learning Model Performance — Actual vs. Predicted ($R^2 = 0.9543$)



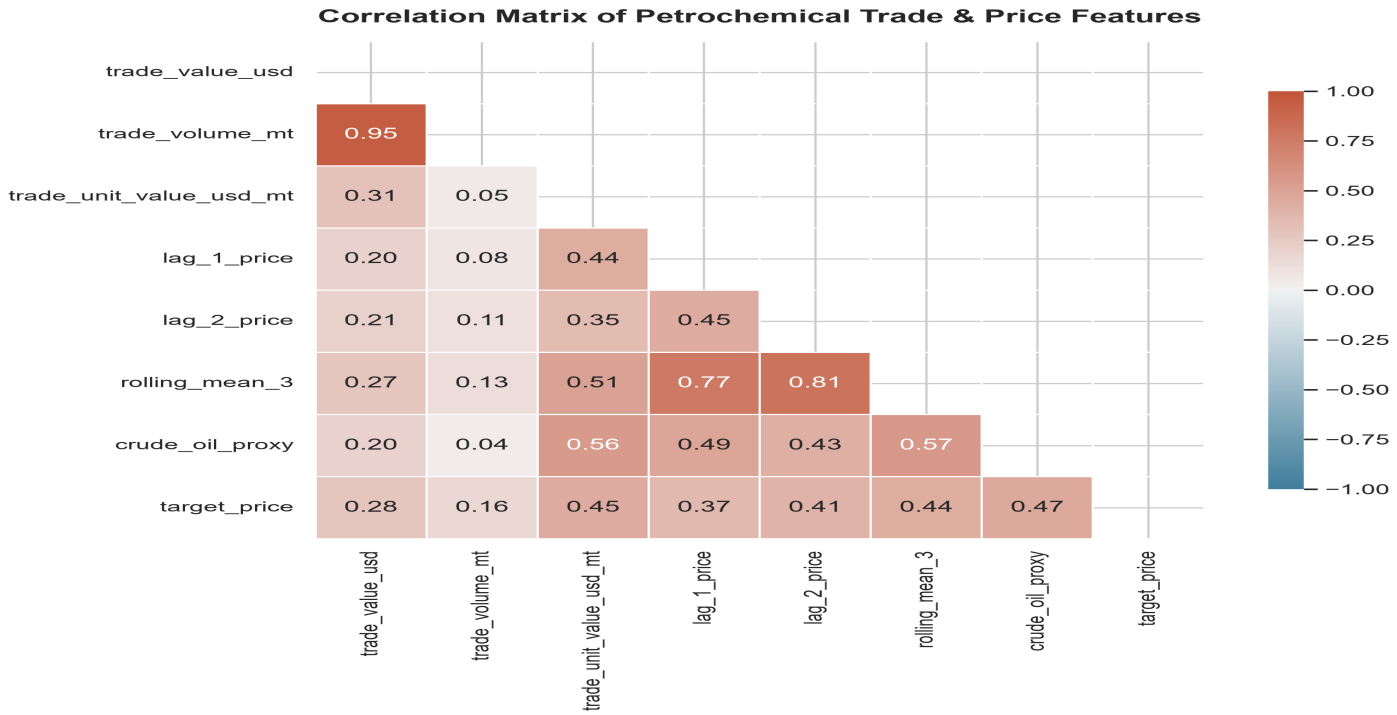
Functional Overview: *Regression diagnostics scatter plot showing predicted versus actual petrochemical trade value across test partitions, demonstrating high explanatory power and low residual dispersion.*

Figure 10: Predictive Feature Importance Ranking



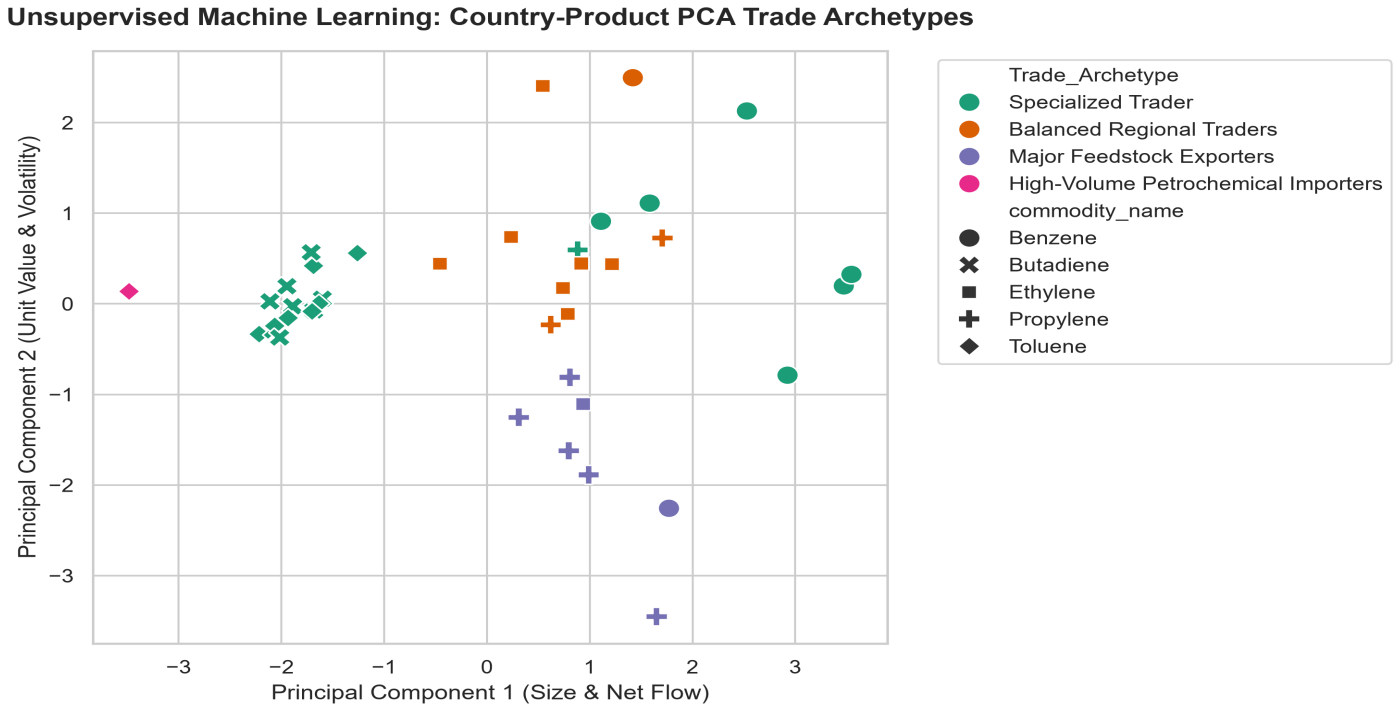
Functional Overview: *Feature attribution analysis demonstrating that cargo volume (MT), crude oil benchmark proxies, and origin/partner country encodings dominate trade value determination.*

Figure 11: Econometric Correlation Heatmap



Functional Overview: *Correlation matrix evaluating dependencies between transaction value, physical volume, unit clearing prices, crude oil proxies, and chronological time indices.*

Figure 12: Unsupervised K-Means Bilateral Corridor Clustering (PCA Biplot)



Functional Overview: *Principal Component Analysis (PCA) biplot displaying K-Means clustering segmentation of international trade routes into high-volume bulk corridors, niche specialty flows, and re-export centers.*